Examples of structurally identical parameter sets to θ_A with varying initial condition weights, fixing c1-c11

In[•]:=
```
cond123solved = Join[{v1 == cv1}, {v2 == cv2}, {v3 == cv3}]
```
Out[•]=
```
{v1 == cv1, v2 == cv2, v3 == cv3}
```

In[•]:=
```
cond456solved = Join[{l1 == cl4}, {l2 == cl5}, {l3 == cl6}]
```
Out[•]=
```
{l1 == cl4, l2 == cl5, l3 == cl6}
```

In[•]:=
```
cond78 =
 {l1 p12 (l2 + l3 - l2 p21) + (l1 - l1 p12 + l2 p21) (l3 - l3 p31) + l2 (l1 - l1 p12 + l3 p31) ==
   c7, l2 l3 p21 v1 + l3 (l2 - l2 p21) p31 v1 + l1 l3 p12 p31 v2 +
    l1 (l3 - l3 p31) v2 + l2 (l1 - l1 p12) p21 v3 + l1 (l2 - l2 p21) v3 == c8}
```
Out[•]=
```
{l1 p12 (l2 + l3 - l2 p21) + (l1 - l1 p12 + l2 p21) (l3 - l3 p31) + l2 (l1 - l1 p12 + l3 p31) ==
  c7, l2 l3 p21 v1 + l3 (l2 - l2 p21) p31 v1 + l1 l3 p12 p31 v2 +
   l1 (l3 - l3 p31) v2 + l2 (l1 - l1 p12) p21 v3 + l1 (l2 - l2 p21) v3 == c8}
```

In[•]:=
```
cond9 = {a1 * v1 + a2 * v2 + (1 - a1 - a2) * v3 == c9}
```
Out[•]=
```
{a1 v1 + a2 v2 + (1 - a1 - a2) v3 == c9}
```

In[•]:=
```
othercond = {a1 * (v1^2) + a2 * (v2^2) + (1 - a1 - a2) * (v3^2) == c10,
                l1 * a1 * (v3 - v1 + (v2 - v3) * p12) + l2 * a2 * (v3 - v2 + (v1 - v3) * p21) +
   l3 * (1 - a1 - a2) * (v2 - v3 + (v1 - v2) * p31) == c11}
```
Out[•]=
```
{a1 v1^2 + a2 v2^2 + (1 - a1 - a2) v3^2 == c10, (1 - a1 - a2) l3 (p31 (v1 - v2) + v2 - v3) +
   a2 l2 (-v2 + p21 (v1 - v3) + v3) + a1 l1 (-v1 + p12 (v2 - v3) + v3) == c11}
```

In[•]:=
```
allcond = Join[cond123solved, cond456solved, cond78, cond9, othercond]
```
Out[•]=
```
{v1 == cv1, v2 == cv2, v3 == cv3, l1 == cl4, l2 == cl5, l3 == cl6,
 l1 p12 (l2 + l3 - l2 p21) + (l1 - l1 p12 + l2 p21) (l3 - l3 p31) + l2 (l1 - l1 p12 + l3 p31) ==
  c7, l2 l3 p21 v1 + l3 (l2 - l2 p21) p31 v1 + l1 l3 p12 p31 v2 +
   l1 (l3 - l3 p31) v2 + l2 (l1 - l1 p12) p21 v3 + l1 (l2 - l2 p21) v3 == c8,
 a1 v1 + a2 v2 + (1 - a1 - a2) v3 == c9, a1 v1^2 + a2 v2^2 + (1 - a1 - a2) v3^2 == c10,
 (1 - a1 - a2) l3 (p31 (v1 - v2) + v2 - v3) +
   a2 l2 (-v2 + p21 (v1 - v3) + v3) + a1 l1 (-v1 + p12 (v2 - v3) + v3) == c11}
```

Apply to the example model A

Example: at steady state

In[•]:=
```
sysAss =
 FullSimplify[allcond /. {v1 → 20, v2 → -15, v3 → 0, l1 → 1, l2 → 1 / 2, l3 → 3 / 10,
    p12 → 1 / 5, p21 → 3 / 10, p31 → 7 / 10, a1 → 237 / 1441, a2 → 24 / 131}]
```

Out[•]=
$$\{cv1 == 20,\ cv2 == -15,\ cv3 == 0,\ cl4 == 1,\ 2\,cl5 == 1,\ 10\,cl6 == 3,\ 2000\,c7 == 1441,\ 100\,c8 == 39,\ 1441\,c9 == 780,\ 1441\,c10 == 154\,200,\ c11 == 0\}$$

In[•]:=
```
solsAss = Solve[sysAss, {cv1, cv2, cv3, cl4, cl5, cl6, c7, c8, c9, c10, c11}][[1]]
```

Out[•]=
$$\left\{cv1 \to 20,\ cv2 \to -15,\ cv3 \to 0,\ cl4 \to 1,\ cl5 \to \frac{1}{2},\ cl6 \to \frac{3}{10},\ c7 \to \frac{1441}{2000},\ c8 \to \frac{39}{100},\ c9 \to \frac{780}{1441},\ c10 \to \frac{154\,200}{1441},\ c11 \to 0\right\}$$

In[•]:=
```
c11val = FullSimplify[othercond[[2]] /. {v1 → 20, v2 → -15, v3 → 0,
    l1 → 1, l2 → 1 / 2, l3 → 3 / 10, a1 → 237 / 1441, a2 → 24 / 131,  c11 → 0}]
```

Out[•]=
$$466 + 237\,p12 == 176\,p21 + 658\,p31$$

In[•]:=
```
solAss = Solve[allcond /. solsAss](* B and A *)
```

Out[•]=

$$\left\{\left\{a1 \to \frac{237}{1441},\ a2 \to \frac{24}{131},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\right.\right.$$
$$\left.p12 \to \frac{66}{395},\ p21 \to \frac{79}{220},\ p31 \to \frac{158}{235},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},$$
$$\left\{a1 \to \frac{237}{1441},\ a2 \to \frac{24}{131},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to \frac{1}{5},\ p21 \to \frac{3}{10},\ p31 \to \frac{7}{10},\right.$$
$$\left.v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},\ \left\{a1 \to \frac{237}{1441},\ a2 \to \frac{24}{131},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to\right.$$
$$\frac{-6\,252\,568\,432 + \frac{31\,239\,118\,723\,\left(23\,955-26\,\sqrt{895}\right)}{50\,435} - \frac{292\,816\,415\,\left(23\,955-26\,\sqrt{895}\right)^2}{14\,535\,367} + \frac{15\,698\,\left(23\,955-26\,\sqrt{895}\right)^3}{72\,676\,835}}{52\,546\,455},$$
$$p21 \to$$
$$\frac{-6\,149\,249\,242 + \frac{31\,093\,230\,253\,\left(23\,955-26\,\sqrt{895}\right)}{50\,435} - \frac{292\,816\,415\,\left(23\,955-26\,\sqrt{895}\right)^2}{14\,535\,367} + \frac{15\,698\,\left(23\,955-26\,\sqrt{895}\right)^3}{72\,676\,835}}{39\,021\,840},$$
$$p31 \to \frac{23\,955 - 26\,\sqrt{895}}{50\,435},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\Big\},$$
$$\left\{a1 \to \frac{237}{1441},\ a2 \to \frac{24}{131},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to\right.$$
$$\frac{-6\,252\,568\,432 + \frac{31\,239\,118\,723\,\left(23\,955+26\,\sqrt{895}\right)}{50\,435} - \frac{292\,816\,415\,\left(23\,955+26\,\sqrt{895}\right)^2}{14\,535\,367} + \frac{15\,698\,\left(23\,955+26\,\sqrt{895}\right)^3}{72\,676\,835}}{52\,546\,455},$$
$$p21 \to$$
$$\frac{-6\,149\,249\,242 + \frac{31\,093\,230\,253\,\left(23\,955+26\,\sqrt{895}\right)}{50\,435} - \frac{292\,816\,415\,\left(23\,955+26\,\sqrt{895}\right)^2}{14\,535\,367} + \frac{15\,698\,\left(23\,955+26\,\sqrt{895}\right)^3}{72\,676\,835}}{39\,021\,840},$$
$$p31 \to \frac{23\,955 + 26\,\sqrt{895}}{50\,435},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\Big\}\Big\}$$

Example: at equal (not at SS)

In[•]:=
```
sys2eq = FullSimplify[allcond /. {v1 → 20, v2 → -15, v3 → 0, l1 → 1, l2 → 1 / 2,
    l3 → 3 / 10, p12 → 1 / 5, p21 → 3 / 10, p31 → 7 / 10, a1 → 1 / 3, a2 → 1 / 3}]
```

Out[•]=

$$\left\{cv1 == 20,\ cv2 == -15,\ cv3 == 0,\ cl4 == 1,\ 2\,cl5 == 1,\ 10\,cl6 == 3,\right.$$
$$\left.2000\,c7 == 1441,\ 100\,c8 == 39,\ 3\,c9 == 5,\ 3\,c10 == 625,\ c11 == -\frac{193}{60}\right\}$$

In[•]:=
```
solkAeq = Solve[sys2eq, {cv1, cv2, cv3, cl4, cl5, cl6, c7, c8, c9, c10, c11}][[1]]
```

Out[•]=

$$\left\{cv1 \to 20,\ cv2 \to -15,\ cv3 \to 0,\ cl4 \to 1,\ cl5 \to \frac{1}{2},\right.$$
$$\left.cl6 \to \frac{3}{10},\ c7 \to \frac{1441}{2000},\ c8 \to \frac{39}{100},\ c9 \to \frac{5}{3},\ c10 \to \frac{625}{3},\ c11 \to -\frac{193}{60}\right\}$$

In[ ]:=
```
sol2Aeq = Solve[allcond /. solkAeq]
```

Out[ ]=

$$\left\{\left\{a1 \to \frac{1}{3},\ a2 \to \frac{1}{3},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to \frac{1}{5},\ p21 \to \frac{3}{10},\ p31 \to \frac{7}{10},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},\ \left\{a1 \to \frac{1}{3},\ a2 \to \frac{1}{3},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to 0.464\ldots,\ p21 \to 1.66\ldots,\ p31 \to -0.219\ldots,\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},\ \left\{a1 \to \frac{1}{3},\ a2 \to \frac{1}{3},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to 0.110\ldots,\ p21 \to 0.197\ldots,\ p31 \to 0.671\ldots,\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},\ \left\{a1 \to \frac{1}{3},\ a2 \to \frac{1}{3},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to 0.816\ldots,\ p21 \to 0.455\ldots,\ p31 \to 1.43\ldots,\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\}\right\}$$

Example: a1 = 1 (not at SS) - solved numerically they are equal

In[ ]:=
```
sysAn1 = FullSimplify[allcond /. {v1 → 20, v2 → -15, v3 → 0, l1 → 1, l2 → 1 / 2,
    l3 → 3 / 10, p12 → 1 / 5, p21 → 3 / 10, p31 → 7 / 10, a1 → 1, a2 → 0}]

solkAn1 = Solve[sysAn1, {cv1, cv2, cv3, cl4, cl5, cl6, c7, c8, c9, c10, c11}][[1]]
```

Out[ ]=

$$\{cv1 == 20,\ cv2 == -15,\ cv3 == 0,\ cl4 == 1,\ 2\,cl5 == 1,\ 10\,cl6 == 3,\ 2000\,c7 == 1441,\ 100\,c8 == 39,\ c9 == 20,\ c10 == 400,\ c11 == -23\}$$

Out[ ]=

$$\left\{cv1 \to 20,\ cv2 \to -15,\ cv3 \to 0,\ cl4 \to 1,\ cl5 \to \frac{1}{2},\ cl6 \to \frac{3}{10},\ c7 \to \frac{1441}{2000},\ c8 \to \frac{39}{100},\ c9 \to 20,\ c10 \to 400,\ c11 \to -23\right\}$$

In[ ]:=
```
solAn1 = Solve[allcond /. solkAn1] (* D and A *)
```

Out[ ]=

$$\left\{\left\{a1 \to 1,\ a2 \to 0,\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to \frac{1}{5},\ p21 \to \frac{47}{50},\ p31 \to \frac{23}{42},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},\ \left\{a1 \to 1,\ a2 \to 0,\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to \frac{1}{5},\ p21 \to \frac{3}{10},\ p31 \to \frac{7}{10},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\}\right\}$$

Example: at a3 = 1 (not at SS) - solved numerically they are equal

In[ ]:=
```
sysAn3 = FullSimplify[allcond /. {v1 → 20, v2 → -15, v3 → 0, l1 → 1,
    l2 → 1 / 2, l3 → 3 / 10, p12 → 1 / 5, p21 → 3 / 10, p31 → 7 / 10, a1 → 0, a2 → 0}]
```

Out[ ]=

$$\{cv1 == 20,\ cv2 == -15,\ cv3 == 0,\ cl4 == 1,\ 2\,cl5 == 1,\ 10\,cl6 == 3,\ 2000\,c7 == 1441,\ 100\,c8 == 39,\ c9 == 0,\ c10 == 0,\ 20\,c11 == 57\}$$

In[ ]:=
```
solkAn3 = Solve[sysAn3, {cv1, cv2, cv3, cl4, cl5, cl6, c7, c8, c9, c10, c11}][[1]]
```

Out[ ]=

$$\left\{cv1 \to 20,\ cv2 \to -15,\ cv3 \to 0,\ cl4 \to 1,\ cl5 \to \frac{1}{2},\right.$$
$$\left. cl6 \to \frac{3}{10},\ c7 \to \frac{1441}{2000},\ c8 \to \frac{39}{100},\ c9 \to 0,\ c10 \to 0,\ c11 \to \frac{57}{20}\right\}$$

In[ ]:=
```
sol2An3 = Solve[allcond /. solkAn3](* E and A *)
```

Out[ ]=

$$\left\{\left\{a1 \to 0,\ a2 \to 0,\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to \frac{87}{700},\ p21 \to \frac{7}{200},\right.\right.$$
$$\left.p31 \to \frac{7}{10},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},\ \left\{a1 \to 0,\ a2 \to 0,\ l1 \to 1,\ l2 \to \frac{1}{2},\right.$$
$$\left.\left. l3 \to \frac{3}{10},\ p12 \to \frac{1}{5},\ p21 \to \frac{3}{10},\ p31 \to \frac{7}{10},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\}\right\}$$

Example with three solutions

In[ ]:=
```
sysAn4 = FullSimplify[allcond /. {v1 → 20, v2 → -15, v3 → 0, l1 → 1, l2 → 1 / 2,
    l3 → 3 / 10, p12 → 1 / 5, p21 → 3 / 10, p31 → 7 / 10, a1 → 7 / 22, a2 → 3 / 22}]
```

Out[ ]=

$$\left\{cv1 == 20,\ cv2 == -15,\ cv3 == 0,\ cl4 == 1,\ 2\,cl5 == 1,\ 10\,cl6 == 3,\right.$$
$$\left. 2000\,c7 == 1441,\ 100\,c8 == 39,\ 22\,c9 == 95,\ 22\,c10 == 3475,\ c11 == -\frac{953}{220}\right\}$$

In[ ]:=
```
solkAn4 = Solve[sysAn4, {cv1, cv2, cv3, cl4, cl5, cl6, c7, c8, c9, c10, c11}][[1]]
```

Out[ ]=

$$\left\{cv1 \to 20,\ cv2 \to -15,\ cv3 \to 0,\ cl4 \to 1,\ cl5 \to \frac{1}{2},\right.$$
$$\left. cl6 \to \frac{3}{10},\ c7 \to \frac{1441}{2000},\ c8 \to \frac{39}{100},\ c9 \to \frac{95}{22},\ c10 \to \frac{3475}{22},\ c11 \to -\frac{953}{220}\right\}$$

In[ ]:=
```
sol2An4 = Solve[allcond /. solkAn4](* D, E and A *)
```

Out[ ]=

$$\left\{\left\{a1 \to \frac{7}{22},\ a2 \to \frac{3}{22},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to \frac{1}{5},\ p21 \to \frac{47}{50},\right.\right.$$
$$\left.p31 \to \frac{23}{42},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},\ \left\{a1 \to \frac{7}{22},\ a2 \to \frac{3}{22},\ l1 \to 1,\ l2 \to \frac{1}{2},\right.$$
$$\left. l3 \to \frac{3}{10},\ p12 \to \frac{87}{700},\ p21 \to \frac{7}{200},\ p31 \to \frac{7}{10},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},$$
$$\left\{a1 \to \frac{7}{22},\ a2 \to \frac{3}{22},\ l1 \to 1,\ l2 \to \frac{1}{2},\ l3 \to \frac{3}{10},\ p12 \to \frac{1}{5},\ p21 \to \frac{3}{10},\ p31 \to \frac{7}{10},\right.$$
$$\left.v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\},\ \left\{a1 \to \frac{7}{22},\ a2 \to \frac{3}{22},\ l1 \to 1,\ l2 \to \frac{1}{2},\right.$$
$$\left.\left. l3 \to \frac{3}{10},\ p12 \to \frac{323}{550},\ p21 \to \frac{79}{220},\ p31 \to \frac{233}{231},\ v1 \to 20,\ v2 \to -15,\ v3 \to 0\right\}\right\}$$

Example with four solutions

```
In[•]:= sysAn5 = FullSimplify[allcond /. {v1 → 20, v2 → -15, v3 → 0, l1 → 1, l2 → 1 / 2,
          l3 → 3 / 10, p12 → 1 / 5, p21 → 3 / 10, p31 → 7 / 10, a1 → 1 / 4, a2 → 3 / 20}]
```

Out[•]=

$$\left\{\text{cv1} == 20,\ \text{cv2} == -15,\ \text{cv3} == 0,\ \text{cl4} == 1,\ 2\,\text{cl5} == 1,\ 10\,\text{cl6} == 3,\right.$$
$$\left.2000\,\text{c7} == 1441,\ 100\,\text{c8} == 39,\ 4\,\text{c9} == 11,\ 4\,\text{c10} == 535,\ \text{c11} == -\frac{493}{200}\right\}$$

```
In[•]:= solkAn5 = Solve[sysAn5, {cv1, cv2, cv3, cl4, cl5, cl6, c7, c8, c9, c10, c11}][[1]]
```

Out[•]=

$$\left\{\text{cv1} \to 20,\ \text{cv2} \to -15,\ \text{cv3} \to 0,\ \text{cl4} \to 1,\ \text{cl5} \to \frac{1}{2},\right.$$
$$\left.\text{cl6} \to \frac{3}{10},\ \text{c7} \to \frac{1441}{2000},\ \text{c8} \to \frac{39}{100},\ \text{c9} \to \frac{11}{4},\ \text{c10} \to \frac{535}{4},\ \text{c11} \to -\frac{493}{200}\right\}$$

```
In[•]:= sol2An5 = FullSimplify[Solve[allcond /. solkAn5]]
```

Out[•]=

$$\left\{\left\{\text{a1} \to \frac{1}{4},\ \text{a2} \to \frac{3}{20},\ \text{l1} \to 1,\ \text{l2} \to \frac{1}{2},\ \text{l3} \to \frac{3}{10},\ \text{p12} \to \frac{1}{5},\ \text{p21} \to \frac{47}{50},\right.\right.$$
$$\left.\text{p31} \to \frac{23}{42},\ \text{v1} \to 20,\ \text{v2} \to -15,\ \text{v3} \to 0\right\},\ \left\{\text{a1} \to \frac{1}{4},\ \text{a2} \to \frac{3}{20},\ \text{l1} \to 1,\ \text{l2} \to \frac{1}{2},\right.$$
$$\left.\text{l3} \to \frac{3}{10},\ \text{p12} \to \frac{1}{5},\ \text{p21} \to \frac{3}{10},\ \text{p31} \to \frac{7}{10},\ \text{v1} \to 20,\ \text{v2} \to -15,\ \text{v3} \to 0\right\},$$
$$\left\{\text{a1} \to \frac{1}{4},\ \text{a2} \to \frac{3}{20},\ \text{l1} \to 1,\ \text{l2} \to \frac{1}{2},\ \text{l3} \to \frac{3}{10},\ \text{p12} \to -\frac{3\left(-135+\sqrt{4001}\right)}{2000},\right.$$
$$\left.\text{p21} \to \frac{997-3\sqrt{4001}}{4000},\ \text{p31} \to \frac{999-\sqrt{4001}}{1400},\ \text{v1} \to 20,\ \text{v2} \to -15,\ \text{v3} \to 0\right\},$$
$$\left\{\text{a1} \to \frac{1}{4},\ \text{a2} \to \frac{3}{20},\ \text{l1} \to 1,\ \text{l2} \to \frac{1}{2},\ \text{l3} \to \frac{3}{10},\ \text{p12} \to \frac{3\left(135+\sqrt{4001}\right)}{2000},\right.$$
$$\left.\left.\text{p21} \to \frac{997+3\sqrt{4001}}{4000},\ \text{p31} \to \frac{999+\sqrt{4001}}{1400},\ \text{v1} \to 20,\ \text{v2} \to -15,\ \text{v3} \to 0\right\}\right\}$$